Specialization: Machine Learning – Supervised Learning

Business Focus: Health Insurance

Tool: Python

SUPERVISED LEARNING – HELPING A HMO PRICE INSURANCE PREMIUMS

Project Learning Opportunities

OptiCare HMO is launching a data-driven pricing system to offer fair and personalized health insurance plans. Using customer data, the company aims to predict medical costs through supervised learning regression models

Tools and Technology to be Used

pandas

matplotlib

scikit learn

Business Overview

OptiCare HMO is a contemporary health insurance company dedicated to providing affordable, customized healthcare coverage throughout Sweden. Utilizing data science and technology, OptiCare creates personalized health plans based on each person's distinct health profile, promoting fairness, transparency, and improved financial planning for both clients and the company. The organization's goal is to enhance access to quality healthcare while improving risk management through smart, data-informed decisions.

OPTICARE

SMART HEALTH COVERAGE, TAILORED TO YOU.

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Case Study Overview

Problem Statement

As OptiCare HMO expands its services, the company seeks to implement a **personalized pricing model** for its health insurance plans. Currently, all insured are charged flat or tiered premiums, which do not account for individual risk factors such as age, **BMI, smoking habits, or family health history**. This limits profitability and may lead to unfair pricing. The goal is to build a supervised learning regression model that accurately **predicts a customer's expected medical expenses based on their personal information**, enabling OptiCare to offer **fair, risk-adjusted insurance premiums**.

Rationale for the Project

(What is the Importance of the project to the business)

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Personalized Pricing

Enables fair, data-driven premium pricing tailored to each customer's risk profile, increasing customer trust and satisfaction

Improved Risk Management

Helps the business better estimate future medical costs and reduce financial uncertainty.

Competitive Advantage

Positions OptiCare as a tech-forward, customer-centric HMO in the health insurance market

Case Study Objectives

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Exploratory Data Analysis

Prepare data for training Machine Learning models

Apply Regression to solve problems

Interpret the output of Regression models

Evaluate Regression models

Data Description

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Column Name	Data Type	Description	Example
age	Integer	Age of the individual in years	29
sex	Categorical	Gender of the individual (male, female)	female
bmi	Float	Body Mass Index – a measure of body fat based on height and weight	27.9
children	Integer	Number of dependent children covered by the insurance	2
smoker	Categorical	Smoking status (yes, no)	yes
region	Categorical	Residential region (southwest, southeast, northwest, northeast)	southeast
charges	Float	Target variable: Total medical costs billed for the insurance	16,847.50

Tech Stack

pandas

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Project Workflow

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STEP 1

Understand the business goal: Predict individual medical costs for personalized premium pricing.

STEP 2

Analyze distributions, handle missing values (if any), encode categoricals, and scale features.

STEP 3

Train regression models (e.g., Linear Regression, Random Forest) using labeled data (charges).

STEP 4

Assess performance using metrics like MAE, RMSE, and R^2. Compare models and tune hyperparameters

STEP 5

Interpret results for pricing strategy, explain key drivers (e.g., smoking, BMI), and guide policy design.

READY TO DELVE IN?